



**DEPARTMENT OF THE AIR FORCE
AIR FORCE LIFE CYCLE MANAGEMENT CENTER
WRIGHT-PATTERSON AIR FORCE BASE OHIO**

MEMORANDUM FOR RECORD

FROM: AFLCMC/ROU
Human Systems Program Office
1981 Monahan Way
Wright-Patterson AFB, OH 45433

SUBJECT: Safe-to-Fly (StF) Test Plan for Commercial and Government Off-the-Shelf (COTS, GOTS) Boots

1. The purpose of this Test Plan is the certification of boots worn by Air Force (AF) aircrew as Safe-to-Fly (StF). Any application of the below test methods to articles, not readily identified as footwear for AF use, is in violation of its designated use. If applied to articles not readily identified as footwear, all risks and consequences associated with its use in the environment become the responsibility of the individual user(s). All test data and results provided to support COTS StF determinations shall be within one year from the date of submission.

2. StF Definition. AF StF boots must conform, at a minimum, to the following:

Note: NSN boots managed by the Defense Logistics Agency (DLA), including those for other services, can be designated AF StF after a review of the technical requirements in the associated Purchase Description (PD) or Military Specification (MILSPEC) and a StF determination made by the Aircrew Performance Materiel Leader, equivalent, or appropriately delegated authority.

a. Footwear (boots) shall abide by the following dimensions for the largest size/width called out in the cited specification or PDs (e.g. Men's 16 EEEE or Women's 11 EE):

- (1) Length: less than or equal to 16 inches
- (2) Height: less than or equal to 11.5 inches
- (3) Width: less than or equal to 5.5 inches

Note: Items measuring outside of these dimensions shall be addressed on a case-by-case basis by the Human Systems Division Program Office (AFLCMC/ROU) and the Aircrew Performance Material Leader, equivalent or appropriately delegated authority.

b. Boot Height.

(1) Female: The height of the finished boot, measured from the bottom of the heel tread to the top of the back of the boot, shall be $9\frac{3}{4} \pm \frac{1}{2}$ inches on size 10RF and shall graduate up and down between sizes and widths.

(2) Male: The height of the finished boot, measured from the bottom of the heel tread to the top of the back of the boot, shall be $10 \pm \frac{1}{2}$ inches on size 10 R and shall graduate up and down between sizes and widths.

c. Boot color is primarily Coyote (498), Legacy Sage Green (1641), Desert Tan (499), or Black; other colors/shades may be allowed by specific approval of the Aircrew Performance Material Leader, equivalent or appropriately delegated authority as required and/or requested.

d. Construction methods include the “Sidewall” or “Cup” construction method along with the Direct Molded System (DMS), Welt, Stitch-down, and Injected molded construction processes. Maximum dimensioning in paragraph 2 above must not be exceeded. Other constructions may be allowed, if appropriate, after being addressed by the Aircrew Performance Material Leader, equivalent or appropriately delegated authority.

e. Flame resistance. The boots shall meet the Flame Resistance Test Four requirements of NFPA 1971, Standard on Protective Ensemble for Structural Firefighting (2007 Edition). The areas of the boot to be tested are in accordance with Figure 8-5.5.2 of NFPA 1971; however, a sixth area is added to include the nylon quarter side panels. The requirement that the boot shall not melt, or drip does not apply, although the boot shall not exhibit any burn-through (Reference: Sections 7-10.1 and 8.5.5). The boot shall be considered to have failed the burn-through test when burn-through is observed through all layers of the quarter panel and lining components. The rubber outsole is not subject to the flame resistance requirements.

f. Upper heat resistance. The boots shall meet the Thermal Conductive Heat Resistance Test One requirements of NFPA 1971, Standard on Protective Ensemble for Structural Firefighting, 2007 Edition (NFPA 1971, Section 8-7).

g. Sole heat resistance. The boots shall meet the Thermal Conductive Heat Resistance Test Two requirements of the NFPA 1971, Standard on Protective Ensemble for Structural Firefighting, 2007 Edition (NFPA 1971, Section 8-8).

h. Water Resistance. The boots shall be tested for their resistance to rain or water spray in accordance with AATCC 127 Water Resistance: Hydrostatic Pressure Testing.

i. Safety Toe options, steel or composite, are allowed in any boot otherwise authorized StF. Compliance with ASTM F2412 and ASTM F2413 Impact (I/75), Compression (C/75), and Electrical Hazard (EH) is required. Maximum dimensioning in paragraph 2 above must not be exceeded.

3. Type Boot temperature ranges and some associated specifications (MILSPEC and PD) are:

a. Hot weather ($>70^{\circ}\text{F}$): StF boots shall comply with the requirements listed in paragraphs 1 and 2, above, and conform to either PD-WNUV-11-01 - HW, ST, (BOOT, HOT WEATHER, INDUSTRIAL) or GL-PD-09-02 - HW, FR, (BOOTS, COMBAT, HOT WEATHER - FLAME RESISTANT) with, or without, a Safety-Toe option. They will not require waterproofing, only water resistance (AATCC TM-22).

b. Temperate Weather ($\geq 32^{\circ}\text{F}$ to 70°F): StF boots shall comply with the requirements listed in paragraphs 1 and 2, above, and conform to the tests as described in Purchase Descriptions PD 77AESG-06-01A or PD 77AESG-06-03A, as appropriate, with or without a Safety-Toe option.

c. Intermediate Cold/Wet Weather (14°F to 32°F): No StF boots have been AF certified in this specific temperature range.

d. Cold weather (-20° to 32°F) or Extreme Cold weather (-20°F to -49°F , or below): StF boots shall comply with the requirements of WNU 99 PD 17007, Service Boots, Cold Weather or GL-PD-19-03A, Boot, Combat, Cold Weather.

Note: The amount and type of insulation used, will be stated by the vendor/manufacture. The “clo” (average insulation) value (ASTM C518 Standard Test Method for Steady State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus) for the type and amount of insulation used will be determined and reported by the vendor/manufacture. The whole boot clo tested on a thermal instrumented foot model in accordance with ASTM F1291 Standard Test Method for Measuring the Thermal Insulation of Clothing Using a Heated Manikin, will be reported by the vendor/manufacture, if determined. Government testing and analysis may be required to determine, or verify, the prediction modeling temperature/time ranges to frostbite for the candidate boots.

e. Cold Weather and Extreme Cold Weather boots may be required to meet, or exceed, any of the requirements for Slip testing in accordance with SATRA Method TM 144. The minimum Coefficient of Friction (CoF) performance required for the below listed surface conditions shall be:

Normal cold weather outer sole boots:	Area	CoF
Wing surface (Hydraulic mode)	Flat	0.24
Rough Ice surface (Frost mode)	Heel	0.09
Quarry Tile (Wet mode)	Forepart	0.58
Polycoated Concrete (5% Sodium Lauryl Sulphate (NaLS) in H ₂ O	Heel	0.02

Anti-Slip, cold weather outer-sole boots:	Area	CoF
Wing surface (Hydraulic mode)	Flat	0.24
	Forepart	0.24
Rough Ice surface (Frost mode)	Heel	0.17
	Forepart	0.24
Quarry Tile (Wet mode) Polycoated Concrete (5% Sodium Lustral Sulphate (NaLS) in H2O	Forepart	0.61
	Heel	0.24

j. Rubber outsoles shall be tested for Cold Flex (@-23° C) and meet requirement of 500% @ 7,500 flexes when tested in accordance with ASTM D-1052 Test Method for measuring Rubber Deterioration – Cut Growth Using Ross Flexing Apparatus.

4. Certificate of Compliance (C of C) from a manufacturer's vendor certifying whole-boot, components, and/or materials meet a PD or MIL-SPEC requirement, in lieu of a specific test report, may be submitted for consideration for acceptance depending on the intended use and established requirements for each boot type. Data supporting the C of C may be requested.

5. The manufacturer shall identify whether the boot model/style submitted for certification meets Berry Amendment (10 U.S.C 2533a) criteria. Only Berry Amendment compliant materials/capabilities or Berry waived materials may be submitted for consideration for acceptance.

6. After the initial, or subsequent three-year StF approval, and at the discretion of the Senior Material Leader representative, vendors may submit a memo stating the boot make/model has not changed in materials, nor processing practices, and is still fully Berry Amendment compliant. Reasons for requiring a full StF submission, with sample boots and all required testing data, versus a memo, are noted discrepancies from the field or other noted or reported discrepancies with the materials and/or processing practices for the boot.

7. This document supersedes all previous versions of this boot StF plan, specifically, AFLCMC/WNU Safe-to-Fly Test Plan for boots, dated 1 August 2022.

8. The Aircrew Performance Material Lead Engineer POC for this StF plan is Mr. Don Fleury at (508) 206-3071 or donald.f.fleury.civ@army.mil.

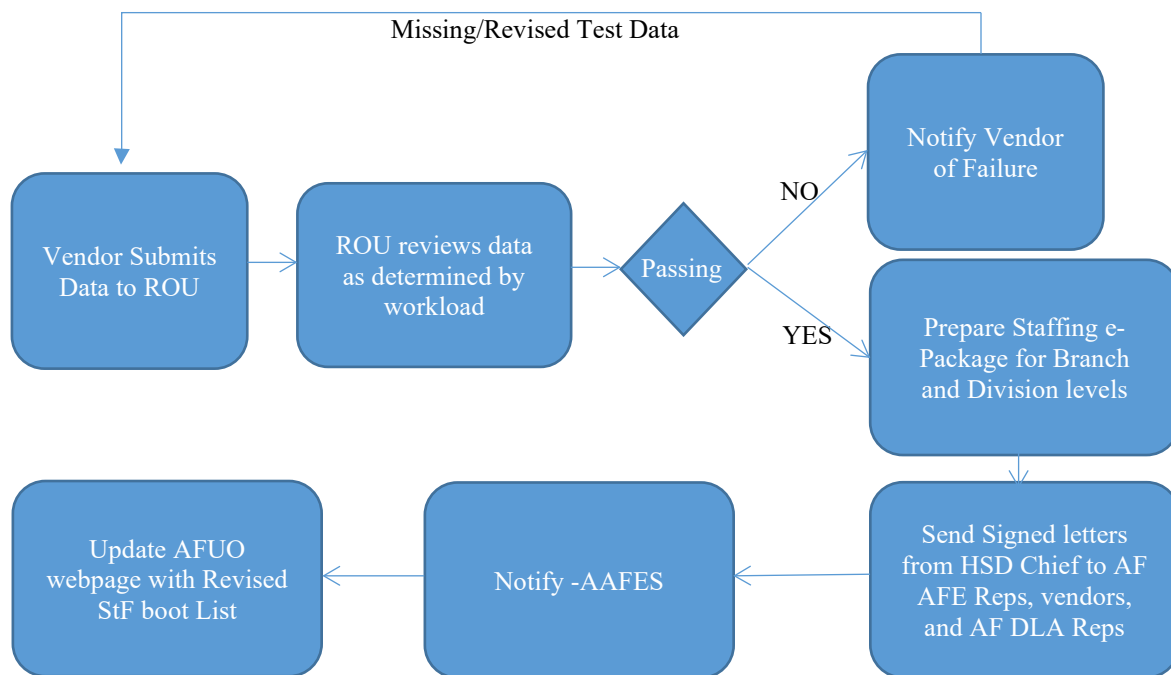
TAYLOR.NICHOL Digitally signed by
E.L.1388626557 TAYLOR.NICHOLE.L.1388626557
Date: 2026.04.08 14:51:22 -04'00'

NICHOLE TAYLOR, NH-04, DAF
Chief Engineer, Human Systems Division
AFPEO Combat Readiness

Attachment:
ROU Process for StF Boots

Attachment
ROU Process for StF COTS Boots

The ROU process for StF COTS boots is outlined in the graphic below. Vendors provide data and boot samples at their convenience. The data (not older than 1-year from submission date) is reviewed and compared to the StF requirements by the assigned engineer(s). A package is prepared that establishes the boots meet all required StF criteria as outlined in the StF plan and coordinated for review and approval. The package includes a boot StF checklist (detailing the requirements and referencing the testing results), testing/pass/fail results, a MAJCOM StF boot memo, a Vendor StF boot memo and a revised AF StF boot list memo.



A vendor StF letter (cc AAFES) is prepared for each specific manufacturer model/style for a three-year period. Multiple boot model/styles may be included in a single memo. The revised StF list is attached to the MAJCOM letter and uploaded on various Aircrew Flight Equipment (AFE) sites. The letters are signed by the Chief of Human System Division or designated representative. Once the package is approved, signed/dated letters are sent to the manufacturer indicating the specific commercial boot style(s) is approved as StF for three years. Note the vendor letter includes a statement to the effect that any material or manufacturing changes will void the StF approval unless the Vendor coordinates the planned change and obtains AF approval from AFLCMC/ROU Materiel Leader and certification package resubmission, beforehand. The MAJCOM StF letter is provided to inform the major AF commands that a new boot has been added to the StF lists or an existing boot has been recertified for an additional three years.

NOTE: New sample boots, supplanted by the recertification process with newer boots, or withdrawn by the vendor for consideration, may be returned to the vendor per their request. Otherwise, they will be disposed of.